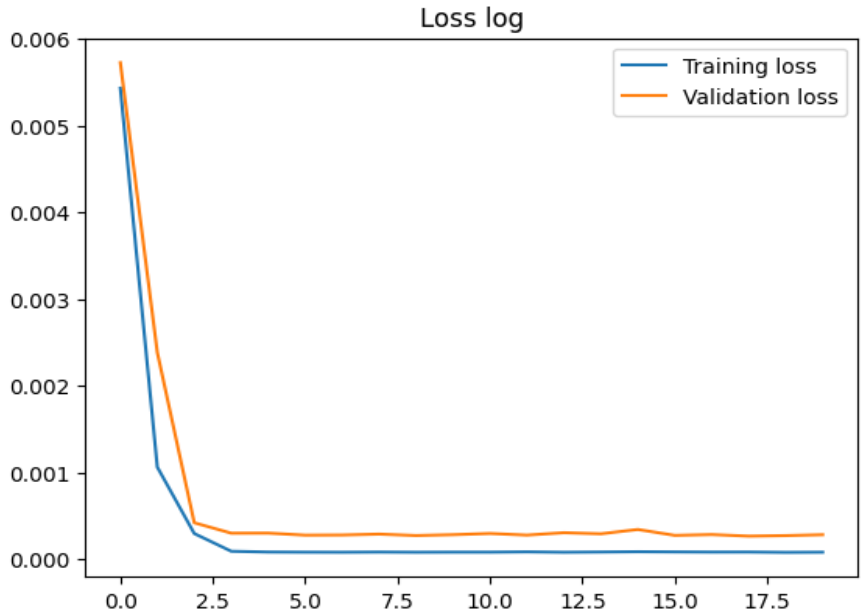
Assignment 1: Part D

Restatement and Summary:

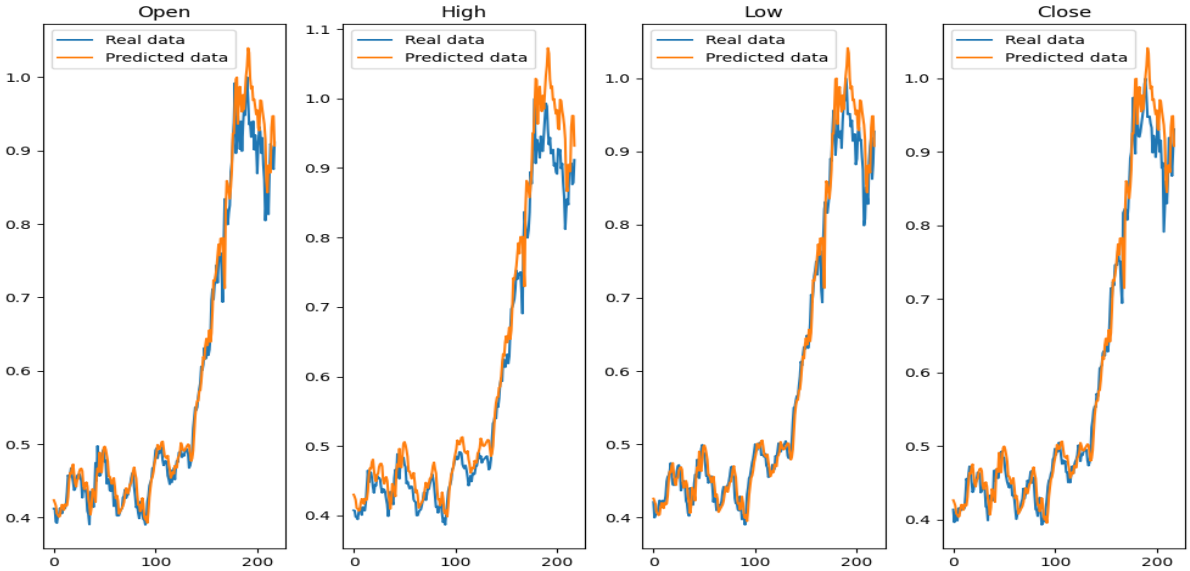
The research focuses on analysing NVIDIA Corporation's stock market dynamics using several key datasets: daily stock price data (including opening, closing, high, and low prices), trading volume data, and event-related data (such as earnings reports and product launches). The primary goal is to understand how trading volume and specific events influence NVIDIA's stock price movements over time. This involves identifying daily and seasonal price trends, analysing volatility, and correlating price changes with significant events. To achieve this, various models, including LSTM and GRU networks, SVM, and CNNs, are employed to handle the temporal and non-linear nature of the data.

The analysis seeks to uncover patterns in price trends, volatility, and the impact of significant events. Data preprocessing steps include handling missing values, scaling data, and transforming features to enhance analysis clarity. LSTM networks are identified as the most suitable model due to their strength in capturing long-term dependencies and non-linear relationships. These models are refined through careful data preparation, hyperparameter tuning, and validation strategies. Performance is assessed using metrics like MAE, RMSE, and R^2. The results provide insights into how trading volume and events affect NVIDIA's stock prices, aiding in strategic decision-making and market understanding.

Analysis and Visualisation:

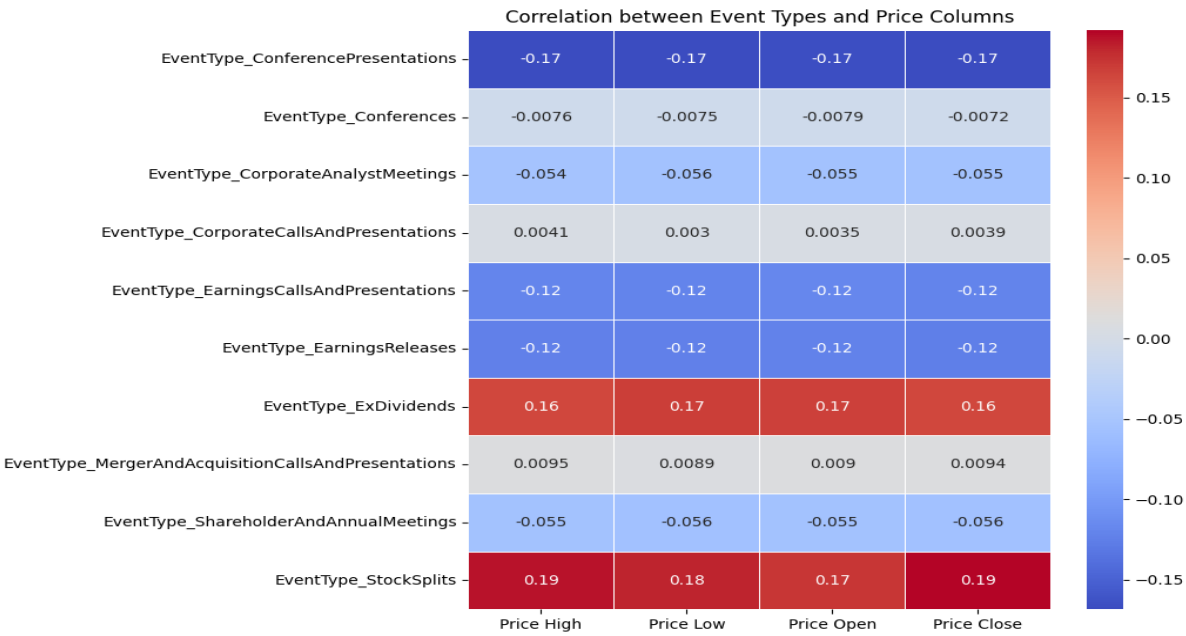


This chart displays the loss during the training and validation phases of a model over several epochs. The x-axis represents the epoch number, while the y-axis represents the loss value. The chart shows two lines: one for training loss (in blue) and one for validation loss (in orange). Initially, both training and validation losses decrease sharply, indicating that the model is learning effectively. After the first few epochs, both losses stabilize at a low value, suggesting that the model is well-trained and not overfitting.



Predicted vs. Real Data

This figure consists of four side-by-side plots, each comparing real and predicted data for different stock price categories: Open, High, Low, and Close prices. The x-axis represents time or the index of data points, and the y-axis represents the normalized price values. The orange line represents the predicted data, while the blue line represents the actual data. The close alignment between the two lines across all four plots suggests that the model's predictions are highly accurate, closely following the real data.



Correlation Heatmap

This heatmap shows the correlation between different event types (listed on the y-axis) and stock price columns (Price High, Price Low, Price Open, Price Close) listed on the x-axis. The colour intensity represents the strength and direction of the correlation: blue shades indicate negative correlations, while red shades indicate positive correlations. For instance, "EventType_ExDividends"

and "EventType_StockSplits" show positive correlations with stock prices, while "EventType_ConferencePresentations" shows a negative correlation. These insights suggest how different events might impact stock prices in positive or negative ways.

The images provide insights into the performance of a model trained to predict stock prices and the influence of various events on these prices. The first image confirms that the model is well-trained, with training and validation losses stabilizing at low values, indicating good performance without overfitting. The second image shows that the model's predictions align closely with the actual stock prices across various categories (Open, High, Low, Close), confirming the model's predictive accuracy. The third image offers a broader analysis of how different corporate events correlate with stock prices, revealing patterns that could be useful for financial analysis and decision-making. For example, events like stock splits and ex-dividends show a positive correlation with stock prices, suggesting potential price increases following these events.

Improvement of Situation:

To enhance outcomes and model performance in analysing NVIDIA's stock market dynamics, several key techniques are crucial.

1. Data Preparation and Preprocessing: Begin by handling missing values through imputation methods like mean, median, or K-Nearest Neighbors (KNN). Normalize or standardize data using Min-Max scaling or standardization to ensure comparability. Aggregate and transform data using techniques like rolling windows to reveal significant patterns.

2. Model Selection and Refinement: Employ Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) networks for capturing temporal dependencies, tuning hyperparameters such as layers, units, and learning rates, and validating through cross-validation or rolling-window methods. Support Vector Machines (SVM) should be fine-tuned for kernel types and regularization parameters, with performance evaluated via cross-validation. Convolutional Neural Networks (CNNs) can be used for feature extraction, requiring adjustments in convolutional layers and kernel sizes, validated through time-based splits.

3. Performance Evaluation and Tuning: Assess model accuracy with metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). Conduct error analysis to identify performance gaps and refine models based on these insights.

Implementing these techniques ensures robust, accurate models that provide valuable insights into the impacts of trading volume and events on NVIDIA's stock prices.

Conclusion and Future Work:

The analysis of NVIDIA's stock market dynamics using advanced models like LSTM networks has provided valuable insights into how trading volume and significant events impact stock price movements. Key findings include the identification of seasonal trends, periods of heightened volatility, and the influence of specific events on price dynamics. The LSTM models, with their ability to capture long-term dependencies and non-linear relationships, demonstrated superior performance in predicting price movements and understanding market behavior.

Next Direction for Research: Future research should explore integrating additional external factors such as macroeconomic indicators and sentiment analysis from news sources to enrich the model's

predictive capabilities. Investigating hybrid models that combine LSTM with attention mechanisms or other advanced neural architectures could further enhance forecasting accuracy.

This study underscores the effectiveness of LSTM networks in capturing complex temporal patterns in stock market data. By refining models and incorporating additional variables, we can gain deeper insights into market trends and improve investment strategies. The results not only provide a robust framework for understanding stock price dynamics but also pave the way for more nuanced financial analyses.

References:

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